Generative Artificial Intelligence (AI) as an Assistant

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Chapter 1

Introduction

Generative artificial intelligence systems are increasingly used to draft, summarize, and explain complex material. They are embedded in everyday tools and often described as assistants, copilots, or collaborators. Despite their rapid adoption, there is little shared understanding of what it means to use generative AI well in research, policy, and analytical contexts.

This book approaches generative AI as an assistant to thinking, not as a source of authority. Rather than treating AI outputs as answers, it treats them as prompts for reflection: ways to surface assumptions, explore alternative framings, clarify arguments, and identify gaps in reasoning or evidence. The central premise is that generative AI can support analytical work without replacing human judgment, responsibility, or methodological care.

In research and policy settings, the risks of uncritical AI use are substantial. Fluent language can obscure uncertainty, fabricated details can appear plausible, and the line between evidence and interpretation can blur. These risks are not failures of individual users alone; they follow directly from how generative models produce text. Recognizing this, the approach taken here emphasizes deliberate use, systematic review, transparency, and epistemic humility.

Throughout this book, generative AI is treated as one tool among many in the analytical process. It can be helpful for organizing ideas, synthesizing material, and improving clarity, but it cannot evaluate evidence, determine truth, assess risk, or assume accountability. Responsibility for interpretation and conclusions always remains with the human analyst or researcher.

This book is written for readers who are curious about generative AI but cautious about its implications: researchers, policy analysts, health system analysts, students, and others engaged in inquiry and investigation. Its goal is not to promote efficiency or automation, but to support thoughtful, disciplined engagement with generative AI as an assistant—used critically, provisionally, and in service of clearer thinking.

1.1 What this book is

This book is a practical and conceptual guide to using generative AI as a research and analysis assistant. It focuses on how large language models can support inquiry and analytical work by helping with:

- Structuring questions and lines of inquiry
- Exploring alternative framings or interpretations
- Summarizing and synthesizing existing material
- Making assumptions, uncertainty, and trade-offs explicit
- Improving clarity in analytical and explanatory writing
- Reviewing and revising AI-assisted drafts responsibly

It emphasizes how AI outputs are evaluated, revised, governed, and integrated into analytical work—not just how they are generated.

1.2 What this book is not

This book does not treat generative AI as:

- A source of factual authority or evidence
- A decision-maker, evaluator, or validator
- A replacement for subject-matter expertise or peer review
- An autonomous research or analysis system

Throughout the book, responsibility for accuracy, interpretation, judgment, and impact remains fully with the human user.

1.3 How to use this book

Readers may engage with this book in different ways:

- Sequentially, as a conceptual guide to AI-assisted research and analysis

- Selectively, by focusing on specific patterns, examples, or review practices
- Reflectively, as a way to examine their own use of generative AI in analytical work

The source files for this book, including all figures and appendices, are maintained in a public repository at this link.

Executive Summary

This book presents a framework for using generative AI as an assistant in policy and health analytics.

1.4 Core principle

Generative AI assists; analysts decide.

AI supports inquiry, drafting, and synthesis, but responsibility for evidence, reasoning, and conclusions remains human.

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Preface

1.5 Why this book

This book was written to support analysts, researchers, and reviewers who are experimenting with generative AI while remaining accountable for evidence, reasoning, and conclusions. It responds to a practical gap: AI tools are increasingly available, but norms for responsible analytical use remain uneven and under-specified.

Rather than offering technical instruction or automation guidance, this book focuses on analytical practice—how to use generative AI as a support for thinking, drafting, and review without displacing judgment or responsibility.

1.6 How this book is different

Many AI guides emphasize efficiency, tooling, or performance. This book takes a different approach. It treats generative AI as an assistant to inquiry, not as a source of authority, evidence, or validation.

The emphasis throughout is on: - deliberate use rather than automation
- review and revision as core analytical disciplines
- explicit handling of uncertainty and assumptions
- clear human authorship and accountability

1.7 How to use this book

This book is designed for selective reading. Different readers may wish to focus on different sections:

- **If you are new to generative AI:**
  Begin with *Conceptual Foundations* and *Prompt Patterns* to understand the analytical stance and basic patterns of use.
- **If you use AI in analytical work:**
  Focus on *Illustrative Examples* and *Reviewing and Revising AI-Assisted Output*, and use **Appendix A** as a practical pre-submission checklist.
- **If you review or govern AI use:**
  Read *Organizational Governance* and *Failure Modes*, alongside **Appendix A**, to understand boundaries, risks, and accountability practices.

1.8 What this book does not attempt

This book does not provide: - tool-specific instructions or vendor guidance
- automation of decisions or analytical conclusions
- legal advice, procurement guidance, or compliance frameworks
- technical evaluation of AI models or training data

Its scope is intentionally limited to analytical judgment and practice.

1.9 Disclaimer

This book reflects general research and analytical practices intended for professional reflection. It does not represent the views, guidance, or policies of any employer, organization, or institution.

Chapter 2

Conceptual Foundations

Generative AI systems are increasingly positioned as assistants in research and investigative work. This book adopts that framing deliberately, while also setting clear limits on what such systems can and cannot do. The central concern is not efficiency or automation, but how AI tools can support thinking without displacing judgment, responsibility, or methodological care.

The figure below summarizes the human-led lifecycle that frames the use of generative AI throughout this book.

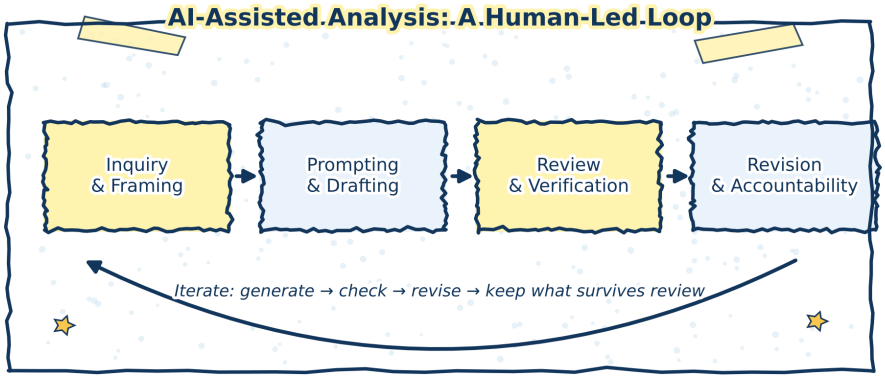

Figure 2.1: AI-assisted analysis as a human-led lifecycle. Generative AI contributes to exploration and drafting, while verification, judgment, and accountability remain human responsibilities.

2.1 Thinking with AI, not delegating judgment

To describe generative AI as a research assistant is to make a normative claim about use. An assistant supports inquiry but does not determine its direction, validate its conclusions, or assume responsibility for its outcomes. In this book, AI systems are treated as instruments that participate in human reasoning processes rather than substitutes for them.

This distinction matters because generative models produce fluent and plausible text without understanding or epistemic awareness. Their outputs can be useful prompts for reflection, comparison, or re-expression, but they do not constitute evidence or justification.

2.2 Capabilities relevant to inquiry

Within these limits, generative AI can support several aspects of research and investigation:

- Structuring complex or ill-defined questions
- Mapping arguments and counterarguments
- Reorganizing or summarizing large volumes of text
- Making assumptions and gaps more visible
- Improving clarity and coherence in analytical writing

These capabilities are best understood as supportive rather than decisive. They shape the space of possible interpretations but do not resolve questions of truth or relevance.

2.3 Epistemic responsibility and accountability

A guiding principle of this book is that epistemic responsibility cannot be outsourced. While AI systems can assist with exploration and drafting, responsibility for accuracy, interpretation, and inference remains human. This stance underpins all subsequent chapters and informs the prompt patterns, examples, and review practices discussed later.

Chapter 3

Prompt Patterns for Inquiry and Synthesis

Prompts are the primary mechanism through which users interact with generative AI systems. In research and analytical contexts, prompts are most effective when they are designed to support inquiry, reflection, and sense-making rather than answer-seeking or decision-making. How a prompt is framed shapes not only what the system produces, but also how its output is interpreted and used.

This chapter treats prompting as a methodological practice rather than a technical trick. The emphasis is on using prompts to open up lines of inquiry, surface alternatives, and make reasoning more explicit—while keeping evaluative judgment and responsibility firmly with the researcher.

The figure below provides a reusable prompt "recipe card" designed for policy and health analytics workflows.

3.1 Prompts as research instruments

A prompt can be understood as a research instrument. Like other instruments, it frames a task, constrains the space of possible responses, and implicitly encodes assumptions about what matters, what is fixed, and what is open to interpretation. Thoughtful prompting therefore requires attention not only to what is being asked, but to what kinds of reasoning are being invited—and what kinds are being excluded.

In this sense, prompts are not neutral. A prompt that asks for a conclusion, recommendation, or "best answer" implicitly shifts authority to the system's output. By contrast, prompts that ask for perspectives, contrasts, alternative

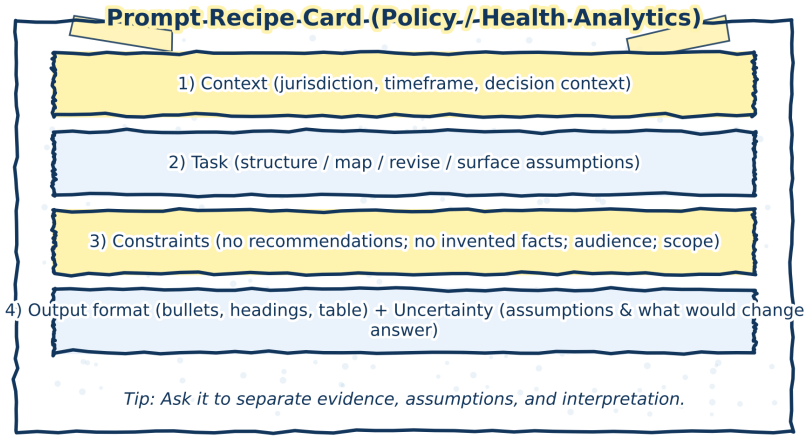

Figure 3.1: Prompt recipe card for inquiry and synthesis. A structured prompt clarifies context, task, constraints, format, and uncertainty without delegating judgment.

framings, or conditional analyses reinforce the exploratory nature of the interaction. They position the AI as a tool for generating material to think with, rather than as a source of determinate answers.

Designing prompts as research instruments supports disciplined inquiry while preserving the researcher's role as the final arbiter of relevance, adequacy, and meaning.

3.2 Core prompt patterns

The following prompt patterns recur throughout this book. They are presented not as templates to be followed mechanically, but as families of approaches that can be adapted to different domains and stages of analysis.

3.2.1 Structuring questions

Prompts can be used to decompose broad, ambiguous, or ill-defined topics into more manageable sub-questions or lines of inquiry. This is particularly useful at early stages of research, when the problem space itself is still being clarified.

The value of this pattern lies in generating candidates for further investigation. The resulting questions should not be treated as exhaustive, definitive, or hierarchically ordered. Instead, they serve as a working map that the researcher can

refine, prune, or reorganize based on domain knowledge and evolving analytical goals.

3.2.1.1 Mini-example: decomposing an ill-defined problem

Hypothetical prompt
"Break this broad research topic into distinct analytical questions that could be explored, without prioritizing or ranking them."

Illustrative outcome
The system proposes several question clusters (e.g. scope, mechanisms, impacts, implementation considerations).

How it is used
The researcher reviews the list, removes irrelevant questions, and reframes others to better align with the project's mandate.

3.2.1.2 Misuse or overreach

Counter-example
Treating the generated questions as a complete or authoritative problem definition and proceeding without further refinement.

This risks locking in the system's framing and prematurely excluding lines of inquiry that require contextual or domain-specific insight.

3.2.1.3 Review and quality-assurance implications

- Assess whether key dimensions are missing or mischaracterized
- Cross-check questions against project objectives and stakeholder needs
- Document how and why the final research questions were selected

3.2.2 Mapping arguments and counterarguments

Generative AI systems can assist in laying out competing positions, interpretations, or lines of reasoning around a topic. When used carefully, this can help clarify the structure of a debate, make tensions explicit, and identify where claims rely on thin evidence or unstated assumptions.

This pattern is most effective when outputs are treated as schematic rather than authoritative. The goal is not to settle disagreements, but to make their contours visible so that they can be evaluated using appropriate evidence, methods, and standards of reasoning.

3.2.2.1 Mini-example: clarifying a contested issue

Hypothetical prompt
"Outline the main arguments and counterarguments surrounding this issue, without assessing which is stronger."

Illustrative outcome
The system presents contrasting positions, each with stated rationales and implied trade-offs.

How it is used
The researcher uses the outline to identify where empirical evidence is required and where normative assumptions are driving disagreement.

3.2.2.2 Misuse or overreach

Counter-example
Presenting the AI-generated argument map as a balanced or complete representation of the debate.

This can obscure power asymmetries, missing perspectives, or the relative credibility of different sources.

3.2.2.3 Review and quality-assurance implications

- Verify that major perspectives are not omitted
- Check that arguments are not falsely symmetrized
- Anchor each position to actual sources or stakeholder views

3.2.3 Surfacing assumptions and gaps

Prompts that ask explicitly about assumptions, dependencies, uncertainties, or missing information can help bring implicit commitments to the surface. These prompts are particularly useful for reviewing drafts, early analytical notes, or preliminary interpretations.

By externalizing assumptions and gaps, this pattern supports reflexivity: it allows researchers to examine how their framing choices shape conclusions, and where additional evidence or clarification may be needed before claims can be responsibly advanced.

3.2.3.1 Mini-example: reviewing a draft analysis

Hypothetical prompt
"Identify key assumptions, uncertainties, and information gaps implicit in this draft."

Illustrative outcome
The system highlights assumptions about scope, data quality, and causal mechanisms.

How it is used
The researcher evaluates which assumptions are acceptable, which require justification, and which indicate the need for further analysis.

3.2.3.2 Misuse or overreach

Counter-example
Treating the AI-identified assumptions as exhaustive and overlooking others that require domain expertise to recognize.

This can create a false sense of methodological completeness.

3.2.3.3 Review and quality-assurance implications

- Compare AI-surfaced assumptions with those identified by peers or reviewers
- Explicitly address key assumptions in the final analysis
- Track how gaps are resolved or acknowledged

3.3 Prompting with constraints and caution

Adding constraints—such as intended audience, scope, uncertainty, or acceptable levels of confidence—helps reinforce the exploratory and provisional nature of AI-assisted work. Constraints do not narrow thinking so much as they discipline it, making explicit the conditions under which an output might be useful.

Throughout this book, prompts are treated as tools for thinking with AI, not as instructions for delegating analytical responsibility. Effective prompting does not eliminate the need for review, verification, or judgment; it makes those activities more focused and more transparent.

Used well, prompt patterns can support clarity without false certainty and exploration without abdication of responsibility. Used uncritically, they risk producing polished outputs that mask unresolved questions. The difference lies not in the prompt itself, but in how its results are interpreted and acted upon.

3.4 Transition: from prompting to evaluation

This chapter has focused on how prompts can be designed to support inquiry, synthesis, and reflexive analysis. The next chapter shifts attention from elicitation to evaluation: how AI-assisted outputs can be reviewed, documented, and governed in ways that preserve accountability and analytical integrity.

Where prompting shapes what is produced, evaluation determines what is accepted, revised, or rejected. Together, these practices form the backbone of responsible AI-assisted research.

Chapter 4

Illustrative Examples of AI-Assisted Work

This chapter presents illustrative examples of how generative AI can be used in research and investigative contexts. The examples are intentionally generic. Their purpose is not to prescribe workflows, endorse specific tools, or define best practices, but to demonstrate patterns of use that align with the conceptual and normative commitments outlined in earlier chapters.

Across all examples, the emphasis is on how AI outputs are interpreted and integrated, rather than on the outputs themselves. The value of AI assistance lies not in producing answers, but in supporting human sense-making, reflection, and revision.

4.1 Early-stage inquiry

In the early stages of a project, the problem space is often under-specified. Questions may be loosely defined, relevant boundaries unclear, and alternative framings insufficiently explored. In this context, generative AI systems can help broaden the space of possible questions, angles, or interpretations.

For example, a researcher might ask for alternative ways of framing a problem, related issues that could warrant attention, or different analytical lenses that might be applied. The resulting outputs function as prompts for further thinking rather than as research findings. They are starting points for exploration, not endpoints for analysis.

4.1.1 Mini-example: exploratory framing

Hypothetical prompt
"List several alternative ways this policy issue could be framed for analysis, highlighting different underlying concerns or objectives."

Illustrative outcome
The system proposes multiple framings (e.g., equity-focused, cost-containment-focused, implementation-focused), each emphasizing different trade-offs.

How it is used
The researcher reviews the list, discards irrelevant framings, and selects one to pursue based on the project's purpose and constraints.

4.1.2 Misuse or overreach

Counter-example
Treating the generated framings as a comprehensive or authoritative taxonomy of the issue, and proceeding as if unlisted framings do not exist.

This overreach risks premature closure and can obscure perspectives that require domain knowledge or stakeholder input to surface.

4.1.3 Review and quality-assurance implications

- Check whether the AI-suggested framings align with the project mandate and research question
- Explicitly document which framings were adopted or rejected, and why
- Use human review to identify missing or mischaracterized dimensions

4.2 Drafting and revising analytical text

Generative AI can assist with drafting and revising analytical text, particularly with respect to clarity, organization, and tone. For instance, it may help identify ambiguous passages, suggest alternative ways of structuring an argument, or offer rephrasings that improve readability for a given audience.

In this role, the system functions as a linguistic and organizational aid rather than as a substantive contributor. Claims, interpretations, and evaluative judgments must remain under human control. AI-assisted revisions should therefore be reviewed not only for surface quality, but for whether they preserve the intended meaning, qualifications, and analytical stance of the original text.

4.2.1 Mini-example: revising for clarity

Hypothetical prompt
"Rewrite this paragraph to improve clarity and flow, while preserving all substantive claims and qualifications."

Illustrative outcome
The system produces a smoother, more concise version with clearer sentence structure.

How it is used
The researcher compares the revision against the original to ensure no claims were strengthened, softened, or reframed unintentionally.

4.2.2 Misuse or overreach

Counter-example
Accepting a revised paragraph without checking whether nuances, uncertainties, or methodological caveats were altered or removed.

This can result in analytically misleading text that appears more confident or definitive than the underlying evidence supports.

4.2.3 Review and quality-assurance implications

- Line-by-line comparison between original and revised text
- Explicit checks for changes in certainty, scope, or emphasis
- Peer or supervisor review for substantive fidelity, not just readability

4.3 Working across multiple sources

When dealing with large bodies of text—such as reports, interview transcripts, policy documents, or literature reviews—generative AI systems can help summarize recurring themes, contrasts, or points of divergence across sources. These

summaries can serve as navigational aids, helping researchers decide where closer reading or deeper analysis is warranted.

However, such outputs should be treated as provisional and incomplete. They do not replace engagement with primary material, nor do they resolve questions about evidentiary weight, methodological quality, or contextual nuance.

4.3.1 Mini-example: cross-document orientation

Hypothetical prompt
"Summarize recurring themes and points of disagreement across these documents, without assessing their validity."

Illustrative outcome
The system highlights several common themes and notes where documents diverge in emphasis or conclusions.

How it is used
The researcher uses the summary to prioritize which documents to read closely and where to look for supporting or contradictory evidence.

4.3.2 Misuse or overreach

Counter-example
Treating the AI-generated summary as a substitute for reading the source material, or citing it as if it reflected evidentiary consensus.

This risks flattening important distinctions and overlooking context, methods, or limitations embedded in the original sources.

4.3.3 Review and quality-assurance implications

- Verify summarized themes against a sample of primary documents
- Use summaries only as orientation tools, not analytical outputs
- Maintain traceability between claims and original sources

4.4 Interpretive control as a unifying principle

Across these examples, a consistent theme is interpretive control. Generative AI can assist with exploration, organization, and expression, but the usefulness of its outputs depends entirely on how they are evaluated and integrated by the researcher.

In each case, the system provides material to think with, not conclusions to accept. Maintaining this distinction is essential for ensuring that AI-assisted work remains analytically rigorous, methodologically transparent, and accountable to the standards of the research context in which it is used.

4.5 Transition: from examples to practice

The examples in this chapter illustrate what AI-assisted work can look like when used in support of inquiry rather than as a substitute for judgment. The next chapter builds on these illustrations by turning to practice: how to review, document, and govern AI-assisted work in ways that make assumptions visible, decisions traceable, and responsibility clear.

Where this chapter focuses on patterns of use, the following chapter addresses patterns of oversight—closing the loop between exploration, interpretation, and accountability.

Chapter 5

Reviewing and Revising AI-Assisted Drafts

Responsible use of generative AI in policy and health analytics requires systematic review and active revision of all AI-assisted outputs. This chapter outlines practices for maintaining analytical rigor, evidentiary discipline, and authorship when a generative AI assistant is used as part of the thinking or writing process.

The central principle is straightforward: AI-generated text is always provisional. It becomes part of an analytical product only through deliberate human evaluation, verification, and integration.

The figure below summarizes a review pipeline that turns AI-assisted drafting into an accountable analytical workflow.

5.1 Review dimensions (what to check, beyond "is it factually correct?")

AI-assisted drafts require review across multiple dimensions. The goal is not only to correct factual errors, but to ensure that reasoning, uncertainty, and accountability remain explicit and defensible.

The categories below describe what to review; Appendix A provides the stage-based checklist describing when these checks typically occur.

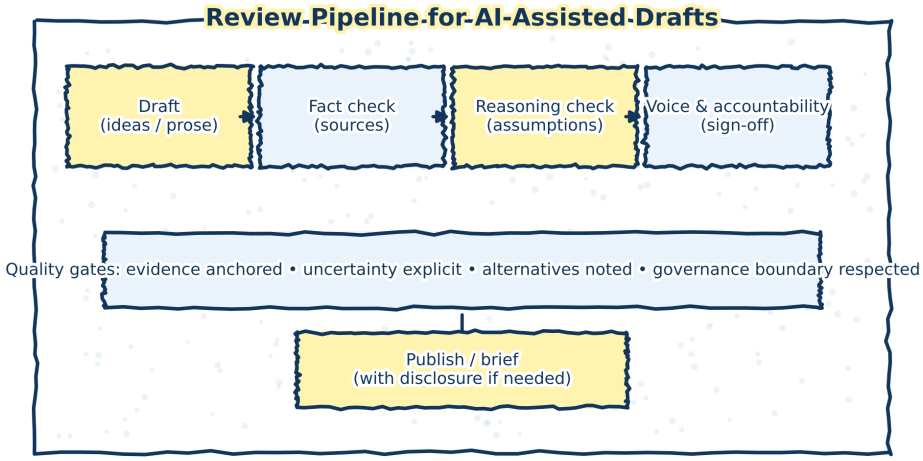

Figure 5.1: Review pipeline for AI-assisted drafts. The workflow emphasizes fact-checking, reasoning review, uncertainty discipline, and explicit accountability before circulation or briefing.

5.1.1 1) Scope and use of AI assistance (governance alignment)

Use is acceptable when AI supports framing, drafting, and reflection without substituting for judgment or validation.

- Ensure AI assistance supported thinking and expression, not decision-making or evaluation.
- Ensure AI output is not treated—explicitly or implicitly—as evidence, validation, or authority.
- Ensure AI use did not substitute for subject-matter expertise, peer review, or institutional oversight.

Risks mitigated: diffusion of responsibility; framing lock-in

5.1.2 2) Framing and problem definition

AI can influence framing early and subtly; review should make framing choices intentional rather than inherited.

- Confirm the framing was chosen intentionally rather than adopted from early AI output.
- Confirm alternative framings were considered or consciously ruled out.

- Confirm the framing does not silently privilege one objective (e.g., efficiency) over others (e.g., equity, feasibility).
- Confirm you can articulate what the framing includes and excludes.

Risks mitigated: framing lock-in; invisible assumptions

5.1.3 3) Factual accuracy and evidence discipline

AI-generated content can be plausible while wrong. Fluency is not evidence.

- Verify all factual claims introduced or modified with AI assistance against primary or authoritative sources.
- Remove fabricated statistics, examples, references, or pseudo-citations.
- Confirm jurisdiction, population, timeframe, and system context are correct and explicit.
- Ensure summaries are not used as substitutes for engagement with source material.

Risks mitigated: fluency masking uncertainty; false balance

5.1.4 4) Reasoning, uncertainty, and analytical integrity

Even when facts are correct, reasoning can be weak, overconfident, or assumption-heavy.

- Confirm conclusions follow logically from premises and evidence.
- Make key assumptions explicit (data, causal, implementation-related).
- Ensure uncertainty, limitations, and evidence gaps are stated clearly and proportionately.
- Ensure claims are not stronger than the supporting evidence allows.
- Distinguish normative judgments from empirical claims.

Risks mitigated: fluency masking uncertainty; invisible assumptions

5.1.5 5) Balance, alternatives, and contested issues

AI may present symmetry where evidence is asymmetric.

- Ensure differences in evidence quality are acknowledged when multiple perspectives are presented.

- Avoid presenting competing views as symmetrical unless analytically justified.
- Ensure marginal or weakly supported positions are not inflated through neutral framing.
- Surface trade-offs and value judgments rather than obscuring them.

Risks mitigated: false balance

5.1.6 6) Authorship, voice, and accountability

AI assistance should not blur responsibility for claims or conclusions.

- Ensure the document reflects a clear human analytical voice consistent with organizational norms.
- Ensure you can defend every claim, framing choice, and conclusion in your own words.
- Ensure responsibility is clearly attributable to the analyst or team.
- Ensure no claim remains solely because "the AI suggested it."

Risks mitigated: diffusion of responsibility

5.1.7 7) Transparency and disclosure (where appropriate)

Transparency supports interpretability and trust.

- Be able to explain how AI assistance shaped the analytical process if asked.
- Ensure required disclosures are accurate, proportionate, and clear.
- Describe AI as supporting functions, not delegated authority.

Risks mitigated: diffusion of responsibility

5.2 Methodological disclaimer: limits of AI-assisted drafting

Generative AI systems generate text based on statistical patterns learned during training and the context provided at use time. They do not have access to authoritative data sources unless explicitly provided, do not evaluate evidence quality, and do not understand policy context, legal constraints, or health system realities.

Practice note
In policy and health analytics, AI outputs should be treated as draft working material, comparable to an unreviewed internal note—not as validated analysis.

Common failure mode
Allowing fluent, confident prose to stand in for evidence-based reasoning or source verification.

5.3 Verifying factual claims

AI-generated text may include inaccuracies, outdated information, or fabricated details. Any factual claim introduced through AI assistance—whether explicit or implied—must be checked against primary sources, administrative data, or authoritative literature.

Fluency, specificity, and technical language are not indicators of reliability.

Practice note (health/policy context)
High-risk factual areas include: - Quantitative claims (rates, trends, effect sizes) - Jurisdictional comparisons - Legal or regulatory descriptions - Causal claims about interventions or policies

Common failure mode
Treating AI-generated background context as "general knowledge" that does not require citation or verification.

The figure below illustrates how to maintain traceability from primary sources to analytical claims in AI-assisted writing.

5.3.1 Review and quality-assurance practices

- Trace each factual claim to a verifiable source
- Replace unsupported claims with sourced statements or explicit uncertainty
- Remove invented examples, citations, or statistics entirely
- Ensure alignment with the correct jurisdiction, timeframe, and population

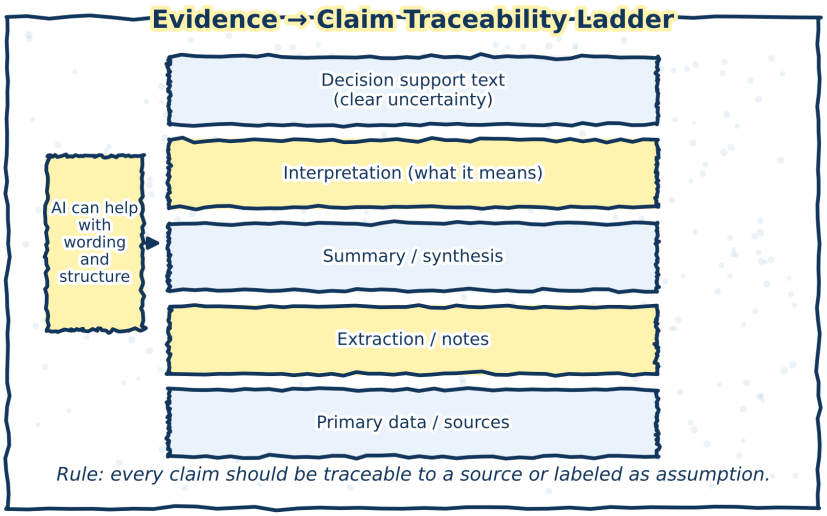

Figure 5.2: Evidence-to-claim traceability ladder. AI may assist with wording and structure, but every claim should be traceable to a source or explicitly labeled as an assumption or uncertainty.

5.4 Evaluating reasoning and analytical structure

Beyond factual accuracy, AI-assisted drafts must be assessed for the quality of reasoning. This includes examining whether conclusions follow from premises, whether alternative interpretations are acknowledged, and whether uncertainty is handled appropriately.

Generative AI systems can reproduce common argumentative structures but do not assess their validity.

Practice note
Ask of any AI-assisted section:
If I removed the prose, would the underlying reasoning still stand on evidence and logic alone?

Common failure mode
Accepting internally consistent reasoning that rests on unexamined assumptions or weak causal logic.

5.4.1 Key review questions

- Are assumptions stated or merely implied?
- Are normative judgments clearly separated from empirical claims?
- Are causal claims proportional to the strength of evidence?
- Are plausible alternatives or counter-interpretations acknowledged?

5.4.2 Review and quality-assurance practices

- Re-outline the argument in bullet form to test logical flow
- Identify where evidence is doing the work vs. where rhetoric is doing the work
- Flag conclusions that appear stronger than the supporting analysis

5.5 Maintaining authorship, accountability, and analytical voice

AI assistance can blur the boundary between drafting and authorship if not handled carefully. In policy and health analytics, maintaining a clear analytical voice is essential for accountability, peer review, and decision-making.

Authorship requires intentional integration: selecting, revising, rejecting, and reshaping AI-generated material so that the final product reflects human judgment and responsibility.

> **Practice note**
> You should be able to explain why every paragraph is written the way it is—regardless of whether AI assisted in drafting it.
>
> **Common failure mode**
> Allowing AI-generated phrasing to dominate tone or framing, resulting in text that feels generic, over-confident, or misaligned with organizational norms.

5.5.1 Review and quality-assurance practices

- Actively rewrite AI-generated text rather than lightly editing it

- Ensure the final voice matches organizational standards and audience expectations
- Confirm that responsibility for conclusions is clearly attributable to the analyst or team

5.6 Transparency and documentation of AI use

In some policy, research, and health analytics contexts, documenting how AI tools were used is appropriate or required. Transparency supports accountability and helps reviewers interpret the analytical process and its limitations.

At a minimum, analysts should be able to explain: - Where AI assistance was used (e.g., scoping, drafting, revision) - What tasks it supported - How outputs were reviewed and validated

> **Practice note**
> Transparency does not require detailing every prompt, but it does require clarity about *where human judgment entered the process.*

> **Common failure mode**
> Treating AI use as either irrelevant ("it was just drafting") or too sensitive to disclose, thereby undermining trust.

5.6.1 Documentation practices

- Maintain brief internal notes on AI-assisted steps
- Be prepared to describe review and validation processes
- Use standardized disclosure language where appropriate

5.7 Iterative use as an analytical discipline

Effective use of generative AI is iterative rather than one-off. Cycles of prompting, reviewing, revising, and discarding outputs help ensure that the system remains an aid to reasoning rather than a source of unexamined conclusions.

Iteration introduces friction, and that friction is a feature—not a flaw—of responsible analytical practice.

Practice note
If AI assistance makes a task feel too easy, that is often a signal to slow down and review more carefully.

Common failure mode
Treating the first plausible output as "good enough" and moving on without systematic review.

5.7.1 Review and quality-assurance practices

- Build explicit review steps into workflows
- Expect to discard or substantially revise most AI-generated text
- Use peer or supervisor review to test robustness and clarity

5.8 Closing principle: AI assists; analysts decide

Across all stages of review and revision, the governing principle remains consistent: generative AI assists, but analysts decide. Responsibility for accuracy, reasoning, interpretation, and impact cannot be delegated.

Used with discipline, AI-assisted drafting can improve clarity and efficiency without compromising rigor. Used without it, the same tools can obscure uncertainty and weaken accountability. The difference lies not in the technology, but in the review practices that surround it.

The review practices described in this chapter are operationalized in Appendix A, which provides a stage-based pre-submission checklist for AI-assisted analytical work.

Chapter 6

Organizational Governance for AI-Assisted Analysis

This chapter outlines governance principles for the responsible use of generative AI assistants in policy and health analytics. Governance here is not understood as technical control or prohibition, but as a set of norms, expectations, and review practices that preserve analytical integrity, accountability, and trust.

Because AI-assisted analysis can influence decisions, resource allocation, and public communication, the question is not whether AI is used, but under what conditions its use remains responsible.

The figure below summarizes the governance boundary that underpins the rest of this chapter.

6.1 Purpose of governance in AI-assisted analysis

In policy and health contexts, analytical work is accountable not only to standards of evidence, but also to institutions, stakeholders, and affected populations. Generative AI systems introduce specific risks—fabrication, over-confidence, framing effects, and obscured assumptions—that governance mechanisms are meant to surface and manage.

> **Practice note**
> Governance does not replace judgment; it makes judgment visible, reviewable, and defensible.

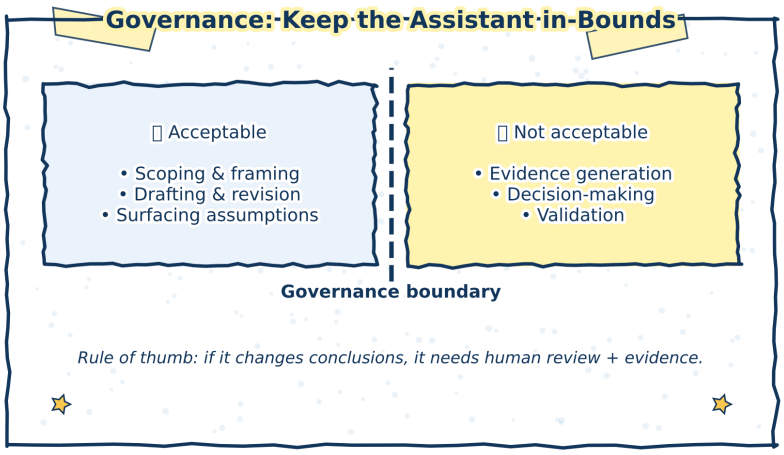

Figure 6.1: Governance boundary for AI-assisted analysis. Generative AI may support framing, drafting, and review, but must not cross into evidence generation, validation, or decision-making.

6.2 Acceptable uses

Generative AI assistance is appropriate when it is used to support analytical thinking and communication, while leaving evaluative judgment with human analysts.

Acceptable uses include:

- **Scoping and question formulation**
  Using AI to help decompose broad or ill-defined problems into candidate analytical questions, dimensions, or lenses—without treating these outputs as definitive or exhaustive.
- **Drafting and revising analytical text**
  Improving clarity, organization, tone, or accessibility of drafts, while retaining human control over claims, interpretations, and conclusions.
- **Surfacing assumptions and gaps**
  Prompting AI to identify implicit assumptions, missing information, or areas of uncertainty in drafts or early analyses, as an aid to reflexive review.

  Illustrative example
  An analyst uses AI to reorganize a briefing note for a senior audience and to flag where causal language appears stronger than the evidence

supports. The analyst revises the text accordingly and verifies all claims against source material.

6.3 Non-acceptable uses

Generative AI assistance is not appropriate where it would substitute for evidence, expertise, or accountability.

Non-acceptable uses include:

- **Generating or validating evidence**
  AI outputs should never be treated as empirical findings, data sources, or confirmation of factual claims.
- **Making or justifying policy decisions**
  AI systems should not be used to recommend options, rank alternatives, or justify decisions, particularly where trade-offs, values, or distributional impacts are involved.
- **Replacing expert or peer review**
  AI assistance does not substitute for subject-matter expertise, methodological review, or institutional oversight.

Common failure mode
Using AI-generated summaries or comparisons as if they represented evidentiary consensus, without engaging with the underlying sources.

6.4 Accountability and authorship

Accountability for AI-assisted work remains fully human. The use of AI does not dilute professional responsibility, institutional accountability, or authorship.

This means that analysts must be able to:

- Explain how AI assistance was used
- Defend analytical choices and conclusions
- Identify and correct errors introduced through AI assistance

Practice note
If an analyst cannot explain or justify a claim without referring to the AI system, that claim does not meet accountability standards.

6.5 Transparency and disclosure

Transparency about AI use supports trust, interpretability, and appropriate review. Disclosure should be proportionate to the role AI played in the work and focused on helping readers understand how conclusions were reached.

In many contexts, this does not require detailed prompt logs, but it does require clarity about:

- Which stages of work involved AI assistance (e.g., scoping, drafting, revision)
- What the system did and did not do
- How outputs were reviewed and validated

Example disclosure (internal)
"Generative AI was used to assist with early drafting and structural revisions. All factual content, analysis, and conclusions were reviewed and validated by the author."

Example disclosure (external, where appropriate)
"AI-assisted tools were used for drafting support; responsibility for analysis and conclusions rests with the authors."

6.6 Governance as a supporting structure

Effective governance does not seek to eliminate AI use, but to embed it within existing analytical standards: evidence review, peer scrutiny, documentation, and accountability.

Used in this way, governance functions as an enabling structure—one that allows generative AI to support analytical work without undermining trust, rigor, or responsibility.

These review practices are operationalized in Appendix A; see Figure.

These governance expectations are reflected in the alignment between review workflow and checklist practices shown in Appendix A Figure

6.6.1 Cross-walk: review dimensions mapped to workflow stages

Review dimension	Stage 1: Draft & framing	Stage 2: Verification	Stage 3: Reasoning & uncertainty	Stage 4: Sign-off & disclosure
Scope & use of AI assistance				
Framing and problem definition			(re-check)	
Factual accuracy & evidence discipline			(claims match strength)	
Reasoning, uncertainty, analytical integrity				(final check)
Balance, alternatives, contested issues				
Authorship, voice, accountability				
Transparency & disclosure				

6.6.2 Governance lens: what reviewers should look for

Governance is strengthened when review makes AI influence visible. The following dimensions help reviewers and managers identify where AI assistance may have crossed boundaries or introduced subtle analytical risks:

- **Use and scope:** Was AI used to support drafting and reflection rather than validation or decision-making?
- **Framing:** Did AI outputs lock in problem framing or obscure alternatives?
- **Evidence discipline:** Are factual claims anchored to sources, with no fabricated details or pseudo-citations?
- **Reasoning and uncertainty:** Are assumptions explicit and confidence proportional to evidence?
- **Balance:** Are contested issues presented with attention to evidence asymmetry?
- **Authorship:** Is accountability clear and attributable?
- **Transparency:** Is AI use explainable and disclosed where appropriate?

Appendix A operationalizes these checks as a stage-based pre-submission checklist.

Chapter 7

Failure Modes in AI-Assisted Policy and Health Analytics

This chapter identifies subtle but consequential failure modes that can arise in AI-assisted analytical work, even when generative AI tools are used with care and good intentions.

These failure modes are not primarily technical errors. They are epistemic and organizational risks that affect how problems are framed, how evidence is interpreted, and how responsibility is exercised—particularly in policy and health analytics, where analysis often informs decisions with real-world consequences.

Understanding these failure modes helps analysts recognize when AI assistance is quietly undermining rigor, clarity, or accountability.

7.1 Why failure modes matter

Generative AI systems produce fluent, plausible text that often aligns with common analytical conventions. This makes their outputs easy to adopt and hard to question. As a result, failures tend to be invisible rather than dramatic: uncertainty is smoothed over, assumptions go unexamined, and responsibility becomes diffuse.

> **Practice note**
> Most AI-related analytical failures arise not from misuse, but from under-review.

7.2 Key failure modes

7.2.1 1. Fluency masking uncertainty

AI-generated text often reads as confident and complete, even when underlying evidence is weak, mixed, or missing. Hedging language may be reduced, caveats removed, or uncertainty expressed less explicitly than in the source material.

Why this matters
In policy and health analytics, tone and confidence can materially influence how findings are interpreted by decision-makers.

> **Illustrative example**
> An AI-revised draft replaces "evidence is limited and mixed" with "evidence suggests," subtly overstating confidence.

Mitigation - Review tone separately from content
- Reintroduce uncertainty explicitly where appropriate
- Compare AI-revised text against original evidence statements

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7.2.2 2. Framing lock-in

Early AI-generated framings—problem definitions, categorizations, or analytical lenses—can silently constrain subsequent analysis. Once adopted, these framings are rarely revisited, even if they exclude relevant perspectives or alternatives.

Why this matters
In policy contexts, framing determines which options appear feasible, relevant, or legitimate.

> **Illustrative example**
> An early AI-generated framing emphasizes efficiency, sidelining equity considerations that were not explicitly prompted.

Mitigation - Generate multiple framings deliberately
- Revisit framing after evidence review
- Ask explicitly: What does this framing exclude?

7.2.3 3. False balance in contested issues

Generative AI often presents opposing views symmetrically, regardless of differences in evidence quality, stakeholder legitimacy, or methodological rigor. This can create an impression of equivalence where none exists.

Why this matters
False balance can distort understanding of consensus, inflate marginal positions, or obscure power and evidence asymmetries.

> **Illustrative example**
> An AI-generated summary presents well-supported public health evidence and anecdotal opposition as parallel "sides" of a debate.

Mitigation - Anchor arguments to evidence strength
- Distinguish empirical claims from normative positions
- Avoid presenting symmetry unless it is analytically justified

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7.2.4 4. Invisible assumptions

AI systems often reproduce common assumptions embedded in training data or prompt context without making them explicit. These assumptions may relate to population homogeneity, data quality, implementation capacity, or causal mechanisms.

Why this matters
Unstated assumptions can drive conclusions while remaining unexamined and unchallenged.

> **Illustrative example**
> An AI-assisted analysis assumes stable baseline trends without noting data gaps or recent system changes.

Mitigation - Prompt explicitly for assumptions and dependencies
- Compare AI-surfaced assumptions with domain-expert review
- Document which assumptions are accepted, tested, or rejected

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7.2.5 5. Diffusion of responsibility

As AI assistance becomes routine, ownership of analytical choices can blur. Decisions about wording, framing, or inclusion may be attributed implicitly to "the system," weakening accountability.

Why this matters
Policy and health analytics require clear lines of responsibility for interpretation and conclusions.

> **Illustrative example**
> A questionable claim is left unchallenged because it "came from the AI" and no one revisits the underlying reasoning.

Mitigation Many of the failure modes described in this chapter arise not from misuse, but from insufficient review; Organizational Governance for AI-Assisted Analysis chapter discusses review and revision as an analytical discipline that mitigates these risks.

- Reinforce authorship and accountability norms
- Require analysts to defend all claims in their own words
- Treat AI output as input, not attribution

7.3 Mitigation as a discipline, not a checklist

Each failure mode requires deliberate analytical discipline, not one-time safeguards. Effective mitigation relies on:

- Systematic review and revision
- Explicit evidence anchoring
- Clear authorship and accountability
- Peer or supervisory scrutiny

These practices are most effective when embedded in routine workflows, rather than applied selectively or after the fact.

The figure below synthesizes the relationship between recurring failure modes and the review practices that mitigate them.

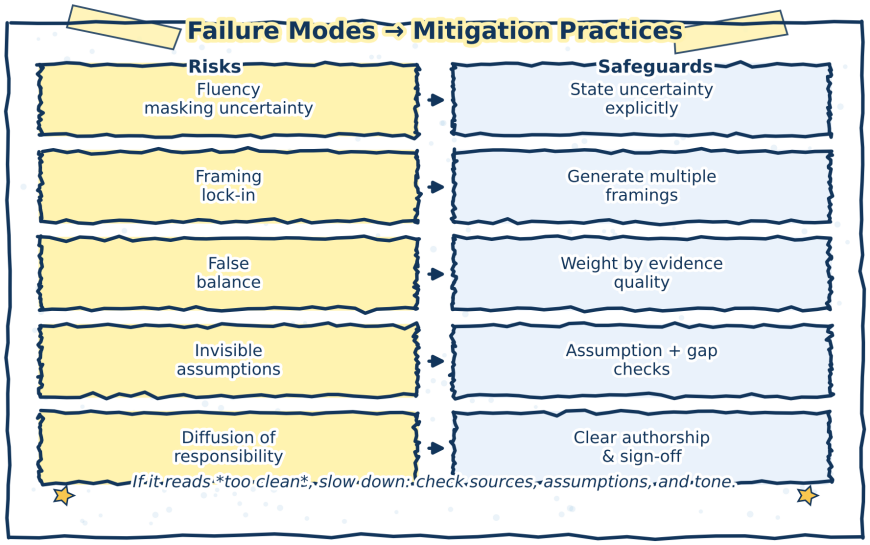

Figure 7.1: Common failure modes in AI-assisted policy and health analytics and corresponding mitigation practices. Most risks arise from under-review rather than misuse.

7.4 Closing reflection

Failure modes in AI-assisted analysis are rarely obvious. They accumulate quietly through small shifts in tone, framing, and responsibility. Recognizing these patterns is a precondition for responsible use.

Used with awareness and discipline, generative AI can support analytical work without compromising rigor. Used without it, the same tools can subtly erode the very standards they are meant to assist.

This work draws on general practices in policy analysis, research methods, and governance rather than any single academic or technical framework.

Additionally, it does not resolve questions of legal liability, procurement policy, or model evaluation. Its focus is on analytical practice and responsibility, not institutional compliance.

Readers interested in broader governance and epistemic discussions may consult the resources listed in Appendix C. For broader discussions of AI governance, epistemic risk, and analytical accountability, readers may wish to consult general literature on:

- Responsible AI and governance frameworks

- Epistemic uncertainty in policy analysis
- Human–AI interaction in knowledge work

Appendix A: Pre-Submission Review Checklist for AI-Assisted Analysis

This checklist supports responsible use of generative AI assistants in policy and health analytics. It is intended for use before circulation, briefing, or publication of any analytical product that involved AI assistance at any stage.

The checklist operationalizes the governance principles and failure modes described in Chapters 6 and 7, and is organized into four stages that mirror the typical lifecycle of AI-assisted analytical work, from early drafting through final sign-off.

The figure below shows how the pre-submission checklist aligns with the review workflow and governance principles described earlier in the book.

Read this figure from top to bottom: the checklist is not an add-on, but a condensed representation of the same review and accountability practices applied throughout the analytical workflow.

7.4.1 Stage 1: Draft & framing

This stage focuses on how the analytical task is framed and how AI assistance is initially used.

- ☐ I can clearly describe where AI assistance was used at this stage (e.g., scoping, outlining, drafting).
- ☐ The problem framing was chosen intentionally and not adopted uncritically from AI output.
- ☐ Reasonable alternative framings were considered or explicitly ruled out.

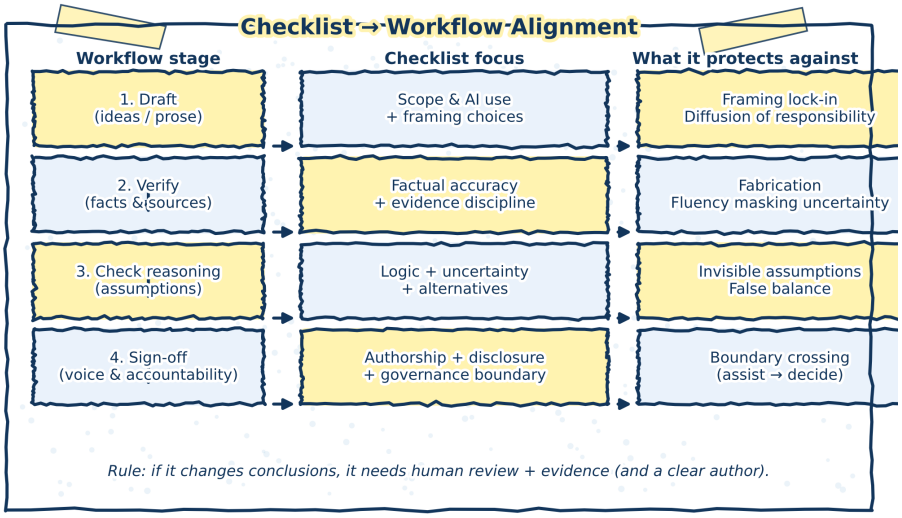

Figure 7.2: Alignment between the pre-submission review checklist, the AI-assisted review pipeline, and governance boundaries.

- ☐ AI assistance supported structuring or expression, not decision-making or evaluation.
- ☐ No AI-generated content is treated as evidence or justification.

7.4.2 Stage 2: Verification

This stage ensures that all factual content introduced or modified with AI assistance is accurate and properly grounded.

- ☐ All factual claims are verified against primary or authoritative sources.
- ☐ No fabricated statistics, examples, citations, or references remain.
- ☐ Jurisdiction, population, timeframe, and system context are correct and explicit.
- ☐ AI-generated summaries are used only as navigational aids, not substitutes for source review.

7.4.3 Stage 3: Reasoning & uncertainty

This stage focuses on analytical integrity, assumptions, and appropriate handling of uncertainty.

- ☐ Conclusions follow logically from stated premises and evidence.
- ☐ Key assumptions (data, causal, or implementation-related) are made explicit.
- ☐ Uncertainty, limitations, and evidence gaps are clearly stated and proportionate.
- ☐ Claims are not stronger or more confident than the underlying evidence supports.
- ☐ Normative judgments are distinguishable from empirical claims.
- ☐ Where multiple perspectives are presented, differences in evidence quality are acknowledged.

7.4.4 Stage 4: Sign-off & disclosure

This stage ensures accountability, authorship, and transparency before circulation or publication.

- ☐ The document reflects a clear human analytical voice consistent with organizational norms.
- ☐ I can explain and defend every claim, framing choice, and conclusion in my own words.
- ☐ Responsibility for the analysis and conclusions is clearly attributable.
- ☐ AI assistance did not cross governance boundaries (e.g., into validation or decision-making).
- ☐ Where appropriate, AI use is disclosed clearly and proportionately. “ —

7.5 Final integrity check

- ☐ If AI assistance were removed entirely, the analysis would still stand on evidence, reasoning, and professional judgment.
- ☐ The use of AI has not reduced clarity about uncertainty, responsibility, or analytical limits.
- ☐ I am comfortable defending this work in a review, audit, or public-facing context.

Governing principle
Generative AI assists; analysts decide.

Appendix B: Glossary of Key Terms

This glossary defines terms as they are used in this book. Definitions are intentionally practical and context-specific, rather than technical or vendor-dependent.

Assistant
A role assigned to a generative AI system in this book. An assistant supports inquiry, drafting, and reflection but does not determine direction, validate conclusions, or assume responsibility for outcomes.

Generative AI system
A software system that produces text or other outputs by generating statistically likely continuations based on patterns learned from data. In this book, generative AI systems are treated as tools for supporting analytical work, not as sources of understanding or authority.

Prompt (as a research instrument)
A structured input used to elicit output from a generative AI system. Like other research instruments, prompts frame a task, constrain possible responses, and encode assumptions. Prompt design shapes the space of outputs but does not confer validity on them.

Evidence vs. interpretation
A distinction between source material (data, documents, empirical findings) and the analytical judgments drawn from them. Generative AI can assist with summarizing or re-expressing evidence, but interpretation and justification remain human responsibilities.

Governance boundary
The conceptual and practical limit beyond which generative AI use becomes inappropriate. In this book, the governance boundary excludes evidence generation, validation, and decision-making, while permitting support for framing, drafting, and review.

Failure mode
A recurring pattern by which AI-assisted analysis can go wrong, often subtly. Failure modes described in this book include fluency masking uncertainty, framing lock-in, false balance, invisible assumptions, and diffusion of responsibility.

Traceability
The ability to link analytical claims back to their sources, assumptions, or explicit judgments. Maintaining traceability ensures that AI-assisted outputs remain interpretable, reviewable, and accountable.

Accountability
The assignment of responsibility for analytical choices, interpretations, and conclusions. In AI-assisted work, accountability always rests with human analysts or teams, regardless of the extent of AI involvement.

Verification
The process of checking factual claims against primary or authoritative sources. Verification is distinct from validation: generative AI outputs require human verification and cannot validate themselves.

Epistemic humility
An orientation toward knowledge that treats conclusions as provisional, uncertainty as explicit, and claims as open to revision. Epistemic humility underpins all responsible uses of generative AI described in this book.

Appendix C: Further Reading

This section lists selected, widely cited resources that discuss themes related to AI governance, epistemic risk, and analytical accountability. These materials are provided for readers who wish to explore the broader landscape; this book does not rely on any single framework or authority and is intended to stand on its own as a guide to analytical practice.

Responsible AI and governance

- Organisation for Economic Co-operation and Development (OECD).
  OECD Principles on Artificial Intelligence.
  https://oecd.ai/en/ai-principles

- Government of Canada.
  Directive on Automated Decision-Making.
  https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai/directive-automated-decision-making.html

- European Commission High-Level Expert Group on Artificial Intelligence.
  Ethics Guidelines for Trustworthy AI.
  https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

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Epistemic uncertainty and policy analysis

- Manski, C. F.
  Public Policy in an Uncertain World: Analysis and Decisions.
  Harvard University Press.

- Saltelli, A., et al.
  Five ways to ensure that models serve society: a manifesto.
  Nature.

Human–AI interaction in knowledge work

- Amershi, S., et al.
  Guidelines for Human–AI Interaction.
  Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI).

- Shneiderman, B.
  Human-Centered Artificial Intelligence.
  Oxford University Press.